

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			$ \begin{array}{c} \text{OH} \\ \\ \text{CH}_3\text{CH}_2\text{C} - \text{CH}_2 - \text{C} \\ \qquad \qquad \quad // \\ \text{H} \qquad \qquad \quad \text{O} \\ \qquad \qquad \quad \backslash \\ \qquad \qquad \quad \text{OH} \end{array} \quad (1) $ balancing i.e. n monomer units and nH₂O molecules on product side (1)			1	2		
	(c)			in condensation polymerisation a small molecule e.g. water is lost; this does not occur with addition polymerisation accept alternatives e.g. different numbers of monomer types, functional groups, atom economy must refer to both types of polymerisation	1	1		1		
	(d)	(i)		surround the flask with a cold water bath	1			1		1
		(ii)		1 mol $\equiv$ 24500 cm³ of hydrogen from 2 mol of the alcohol 184 cm³ from $184 \times 2 / 24500 = 0.015$ mol of the alcohol (1) $M_r = 0.900 / 0.015 = 60$ (1)		2		2	1	2
		(iii)		$ \begin{array}{c} \text{H}_3\text{C} \quad \text{H} \\ \backslash \quad / \\ \text{C} \\ / \quad \backslash \\ \text{H}_3\text{C} \quad \text{OH} \end{array} \quad \text{ecf possible from incorrect } M_r \text{ in part (ii)} $		1		1		1
				Question 11 total	5	8	3	16	3	6